REPORT 1 — ADDENDUM B

Models, Automatability & Tech Stack

LLM vs VLM • the full automatable-criteria map • the build stack for every tier

Factual answers to the team's three follow-up questions on Report 1 and Addendum A: which capabilities need vision (VLM) vs text (LLM) only, exactly which WCAG 2.2 criteria can be automated for testing, and the concrete technology stack — classical ML, generative AI, and agentic AI — to build the POC, Phase 2, and the human-gated automatable checks.

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Responds to: Team questions 1–3 on Report 1 & Addendum A
Grounded in: Report 1 §5, Addendum A, WCAG 2.2 (W3C)
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1. Q1 — LLM only, or VLM? What the POC uses, capability by capability

1.1 The direct answer

> **The POC uses a VLM (Vision-Language Model), not text-only LLMs.** This is stated in Report 1 §5.1 (layer ③b: "VLM: Claude / GPT-4o / Gemini (vision)") and §5.2 ("hosted frontier VLM"). The reason is concrete: two of our four AI checkers must *see the rendered pixels* to do their job, which a text-only LLM cannot do. And critically — **no, the capabilities cannot all be done with LLMs only.** The full system is three different kinds of engine, and "LLM-only" would fail on both ends of it.

Quick definitions (so the distinction is exact)

- **LLM (Large Language Model)** — text in, text out. It can reason over the HTML/DOM and the accessibility tree (both are text), but it is *blind*: it cannot look at the rendered page.
- **VLM (Vision-Language Model)** — a multimodal model that takes *images + text*. It can do everything an LLM does *plus* reason over screenshots. A VLM is a superset of an LLM.
- **Neither is the rule engine.** axe-core and contrast/geometry maths are *deterministic code* — not an LLM at all. "LLM-only" would throw away the reliable, zero-variance layer that catches ~14 criteria.

1.2 Capability-by-capability: does it need vision (VLM), or is LLM-only enough?

DETERMINISTIC no LLM — code LLM-OK text-only suffices VLM needs vision

POC capability	Criteria	Engine needed	Can an LLM alone do it? — why
Rule engine (contrast, target size, reflow, lang, landmarks…)	14 rule	DETERMINISTIC	**No LLM at all.** These are maths/geometry/attribute checks — deterministic code (axe-core). An LLM here would be slower, costlier, and non-reproducible.
Meaningful alt text	1.1.1	VLM	**No — needs vision.** To judge whether alt text describes the image, you must *see* the image. An LLM (blind) can only flag junk patterns like `alt="IMG_1234"` — a weak proxy, not the real check.
Colour-only meaning	1.4.1	VLM	**No — needs vision.** Detecting "information carried by colour alone" (e.g., red = error, with no text/icon) requires looking at the rendered page. LLM-only cannot.
Fake semantics (div-as-button, fake headings, missing role)	1.3.1, 4.1.2	LLM-OK VLM+	**Mostly yes with an LLM** over the DOM + accessibility tree (both text). But the strongest cases — "this *looks* like a heading/button visually but isn't coded as one" — need the VLM to compare the visual render to the code. So LLM-good, VLM-better.
Label correctness (visible label in accessible name)	2.5.3, 3.3.2	LLM-OK	**Yes — often without any LLM.** It is mostly a string/DOM comparison of the visible text vs the computed accessible name; an LLM adds nuance. Vision is only needed if the visible label is itself an image of text.
Headings/labels descriptive (POC-adjacent)	2.4.6	LLM-OK	**Yes.** Judging whether a heading/label is descriptive is a text task over the DOM — LLM-only is sufficient.

1.3 The honest architecture picture

> The POC is deliberately **three engines, not one**: deterministic code for maths/geometry (the rule layer), LLMs for text/semantic judgment over the DOM and accessibility tree, and VLMs for anything that requires seeing the rendered page. Because a VLM is a superset of an LLM, we standardise on a **VLM as the primary model** and route the text-only checks (labels, descriptiveness, much of fake-semantics) to a **cheaper text LLM via the model gateway** to save cost. So: we *use* a VLM; we *also* use LLMs for the text-only parts; and we never try to make an LLM do the deterministic or the visual work.

Why “LLM-only” would fail: it would lose the deterministic layer (14 criteria, zero-variance) at the bottom, and it would be blind to the two vision-dependent checkers (1.1.1, 1.4.1) at the top — roughly a third of the POC’s value gone. The VLM+LLM+rules split is what makes the system both accurate and cost-efficient.

2. Q2 — Every WCAG 2.2 A/AA criterion that can be automated for testing

"Automatable for testing" means a machine can make or strongly assist the determination — with a human confirming the uncertain cases. It is a wider set than "fully decided by rules," and much wider than what the POC ships. Here is the complete picture across all 55 Level A/AA criteria.

2.1 The automatability tiers

TIER 1 — POC TIER 2 — Phase 2 (AI + agentic, human-gated) TIER 3 — Human-led (some partial auto-flagging)

Tier	Count	What "automated" means here
TIER 1 — POC	**20**	Determined automatically in the pull request now: 14 deterministic + 6 AI checkers.
TIER 2 — Phase 2	**20**	Automatable with the fuller AI + agentic layer, human-gated: 5 more AI-primary + 15 interactive/cross-page (agentic) criteria.
TIER 3 — Human-led	**15**	Human judgment/sign-off required; of these, ~6 still get *partial* automated flagging (see 2.3). ~9 have little-to-no automation.
Automatable for testing (Tier 1 + Tier 2)		**40 of 55 (~73%)** under automated/agentic coverage, human-gated — plus partial flagging on ~6 more.

> **Read this precisely:** "40 of 55 automatable for testing" is *not* "40 criteria we can certify." It means automation can test or strongly assist those 40, with humans confirming the uncertain ones. Deterministic-alone is still only ~14 (~25%); the jump to ~73% comes entirely from AI + agentic testing with human gating. Full conformance certification always needs human sign-off.

2.2 The complete lists (what's in each tier)

TIER 1 — POC (20): automated now	
Rule (14)	1.4.2, 1.3.4, 1.3.5, 1.4.3, 1.4.4, 1.4.10, 1.4.11, 1.4.12, 2.4.1, 2.4.2, 2.5.8, 3.1.1, 3.1.2, 3.3.2
AI checkers (6)	1.1.1 (alt), 1.3.1 & 4.1.2 (fake semantics), 1.4.1 (colour), 2.5.3 (label), 2.4.6 (heading/label descriptive)
TIER 2 — Phase 2 (20): automatable, human-gated	
More AI-primary (5)	1.3.2 (meaningful sequence), 1.3.3 (sensory characteristics), 1.4.5 (images of text), 2.4.4 (link purpose), 2.4.7 (focus visible)
Agentic (15)	2.1.1 Keyboard, 2.1.2 No Keyboard Trap, 2.2.2 Pause/Stop/Hide, 2.4.3 Focus Order, 2.4.11 Focus Not Obscured, 1.4.13 Content on Hover/Focus, 3.2.1 On Focus, 3.2.2 On Input, 3.2.3 Consistent Navigation, 3.2.4 Consistent Identification, 3.2.6 Consistent Help, 3.3.1 Error Identification, 3.3.3 Error Suggestion, 3.3.8 Accessible Authentication, 4.1.3 Status Messages
TIER 3 — Human-led (15)	
Partial auto-flag (~6)	1.2.1, 1.2.2, 1.2.3, 1.2.4, 1.2.5 (media — detect missing track / ASR sanity-check), 2.3.1 (flash — CV analysis)
Minimal automation (~9)	2.1.4 Char Key Shortcuts, 2.2.1 Timing Adjustable, 2.5.1 Pointer Gestures, 2.5.2 Pointer Cancellation, 2.5.4 Motion Actuation, 2.5.7 Dragging Movements, 2.4.5 Multiple Ways, 3.3.4 Error Prevention (Legal/Fin), 3.3.7 Redundant Entry

Note: media criteria (1.2.x) and flashing (2.3.1) get *partial* automation (a machine can flag a missing caption track or detect a flash rate), but the conformance decision — are the captions accurate, is the audio description adequate — stays human. Tier boundaries at 2.4.6 (POC vs Phase-2) and 2.3.1 (auto-flag vs human) are the few judgment calls; everything else is firm.

2.3 So, “what else can be automated beyond POC + Phase 2?”

> Beyond the 40 in Tiers 1–2, the honest answer is: **~6 more criteria can be *partially* automated as flagging** (the media set 1.2.1–1.2.5 via missing-track detection + ASR, and 2.3.1 flashing via computer-vision analysis) — but these remain human-decided. The final **~9 criteria** (pointer/gesture/motion alternatives, timing, multiple-ways, legal error-prevention, redundant entry) depend on interaction design, process, and human judgment, and are not meaningfully automatable for a conformance verdict. That is the complete map: **~14 deterministic → ~25 with AI → ~40 with agentic → ~46 with partial flagging → 55 with human sign-off.**

3. Q3 — The tech stack to build every tier

The concrete build stack — classical ML, generative AI, agentic AI, and supporting infrastructure — organised by tier. A shared foundation underpins all three (per Report 2), so components are built once and reused.

3.1 Shared foundation (used by every tier)

Component	Technology (with alternatives)
Render & capture	**Playwright + headless Chromium**; Chrome DevTools Protocol for the **accessibility tree**; full-page + element **screenshots**; DOM snapshot. (alt: Puppeteer, Selenium)
Model gateway	**LiteLLM**-style router in front of all models (VLM / LLM / ASR / embeddings); temperature-0, version-pinned; hosted or self-hosted routing; cost + rate-limit control.
Grounding (RAG)	WCAG 2.2 + **ACT Rules** corpus; **embeddings** (OpenAI `text-embedding-3` or open `bge` / `e5`); **vector DB** (pgvector / FAISS).
Structured output & guardrails	**JSON schema + Pydantic** validation; confidence calibration; "uncertain → human" gating.
Evaluation & MLOps	Golden-set store; scoring harness (precision/recall/F1); CI regression; telemetry & model+prompt audit log; **human-review console** (web UI).
Orchestration & delivery	**GitHub Actions / GitLab CI**; Docker; PR review-comment bot; **SARIF** export; optional VS Code extension.

3.2 Tier 1 — POC build stack

Capability	AI/ML type	Stack
Rule engine (14 criteria)	Deterministic	**axe-core** (WCAG 2.2 ruleset). Contrast = relative-luminance maths; target size/reflow/spacing = geometry + render-and-measure. No ML. (alt: IBM Equal Access, Pa11y)
Meaningful alt (1.1.1), colour-only (1.4.1)	VLM (GenAI)	Frontier **VLM** (Claude / GPT-4o / Gemini vision) on element screenshot + context; RAG-grounded prompt; JSON verdict + confidence. (alt: self-hosted Qwen-VL / Llama-Vision)
Fake semantics (1.3.1, 4.1.2)	LLM +VLM	**LLM** over DOM + accessibility tree; **VLM** for visual-vs-code mismatch; `dom-accessibility-api` to compute role/name.
Label (2.5.3, 3.3.2), headings (2.4.6)	LLM / string	Accessible-name computation + **string similarity**; small **LLM** for nuance/descriptiveness. Cheapest checks — routed to a small model via the gateway.

3.3 Tier 2 — Phase 2 build stack (the new capabilities)

Capability group	AI/ML type	Stack
Images of text (1.4.5)	VLM + OCR	**OCR** (Tesseract / PaddleOCR) + VLM to classify real-text-vs-image-of-text.
Meaningful sequence (1.3.2)	VLM + CV	Bounding-box **reading-order heuristics** (geometry) + VLM to compare visual vs DOM order.
Sensory characteristics (1.3.3), link purpose (2.4.4)	LLM (NLP)	**LLM** text analysis over instructions/link text.
Focus visible (2.4.7)	VLM + CV	

		Screenshot diff (SSIM / pixel diff) of focused vs unfocused state + VLM to confirm a visible indicator.
Interactive/agentic (2.1.1, 2.1.2, 2.4.3, 2.4.11, 1.4.13, 3.2.1, 3.2.2, 2.2.2)	Agentic AI	**LLM/VLM agent with tool-use** (computer-use style) orchestrated by **LangGraph**/custom state machine, driving **Playwright**: keyboard emulation (Tab/Enter), focus tracking via CDP, **DOM MutationObserver** for on-focus/on-input changes; grounded via **MCP**.
Cross-page consistency (3.2.3, 3.2.4, 3.2.6)	Embeddings + LLM	**Embedding similarity** + structural DOM comparison of nav/ components across pages; LLM to judge equivalence.
Error & auth flows (3.3.1, 3.3.3, 3.3.8)	Agentic + LLM	Agent submits invalid forms / walks login; LLM checks error text; VLM/LLM flags cognitive-test CAPTCHAs.
Status messages (4.1.3)	DOM + agentic	**MutationObserver** on `aria-live` regions during agent interaction; assert announcements fire.

3.4 Tier 3 — Human-led, with partial automation assist

Capability	AI/ML type	Stack (automation assists; human decides)
Media: captions/AD/transcripts (1.2.1–1.2.5)	ASR + VLM	Detect missing `<track>` (rule); **ASR (Whisper)** to verify captions exist & roughly match audio; VLM on keyframes to flag likely need for audio description. **Accuracy/adequacy judged by a human.**
Flashing (2.3.1)	Classical CV	**Flash-analysis algorithm** (PEAT-style luminance & red-flash thresholds over rendered frames). Detection automatable; human confirms.
Pointer/motion/timing/process (2.1.4, 2.2.1, 2.5.1/2/4/7, 2.4.5, 3.3.4, 3.3.7)	Static heuristics	Static code heuristics can *flag candidates* (detect drag/motion listeners, time-limit code, submit flows), but the conformance verdict is **human/process** judgment.

3.5 The capability glossary (every AI/ML type, and where it's used)

Capability	Used for
Deterministic rule engines (axe-core)	Contrast, geometry, attributes, structure — the 14 rule criteria. No ML; reproducible.
Classical computer vision	Contrast luminance, target-size geometry, focus-visible screenshot diff (SSIM), flash analysis (PEAT).
OCR	Detecting text baked into images (1.4.5); reading text in image-labels.
LLMs (text, generative)	Semantic judgment over DOM / accessibility tree / text: fake semantics, label/heading/link descriptiveness, sensory-characteristic instructions, error-text checks.
VLMs (multimodal, generative)	Anything needing pixels: meaningful alt, colour-only meaning, focus-visible confirmation, images-of-text, visual-vs-code mismatch.
ASR (Whisper)	Verifying captions/transcripts against audio for the media criteria.
Embeddings + vector search	RAG grounding in WCAG/ACT; cross-page consistency similarity (3.2.3/3.2.4/3.2.6).
RAG	Grounding every AI verdict in the exact WCAG success-criterion text (anti-hallucination).
Agentic AI (tool-use / computer-use)	Interactive testing — keyboard journeys, focus order, on-focus/input, forms, auth, dynamic status.
	Measuring, routing, calibrating, and auditing all of the above.

MLOps (eval harness, gateway, guardrails, telemetry)	

> **One-line summary for the team:** build it as **deterministic code + LLMs + VLMs + ASR + agentic AI** on a shared RAG-grounded, human-gated foundation — not as "an LLM." The POC needs the VLM + LLM + rule layers; Phase 2 adds the agentic layer and more VLM/OCR/CV; the human-gated tier adds ASR and classical CV for partial assist. Each engine is matched to the kind of judgment its criteria actually require.

References

- Report 1 §5 (reference architecture — VLM layer, model gateway, RAG) and Addendum A (the 55-criterion Rule/AI/Human mapping).
- W3C — WCAG 2.2 & *Understanding* docs: w3.org/TR/WCAG22; ACT Rules: w3.org/WAI/…/act/rules.
- Tooling: axe-core github.com/dequelabs/axe-core; Playwright playwright.dev; OpenAI Whisper (ASR) github.com/openai/whisper; PEAT flash-analysis (Trace Center) trace.umd.edu/peat; Model Context Protocol modelcontextprotocol.io.

Accuracy note: the model/engine assignments follow directly from what each WCAG criterion requires (per Addendum A) and Report 1's architecture. Tier counts (20 / 20 / 15) use the single-primary-owner method from Addendum A; the two borderline placements (2.4.6 POC-vs-Phase-2; 2.3.1 auto-flag-vs-human) are noted in-line. Specific products (Claude/GPT-4o/Gemini, Whisper, PEAT, LangGraph) are current examples and are swappable via the model gateway.